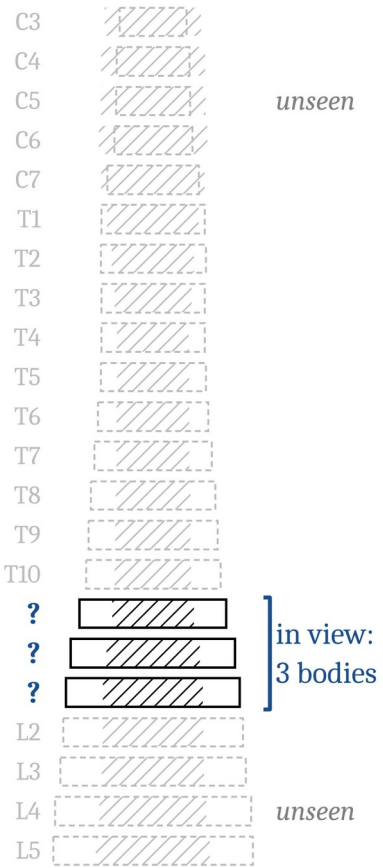
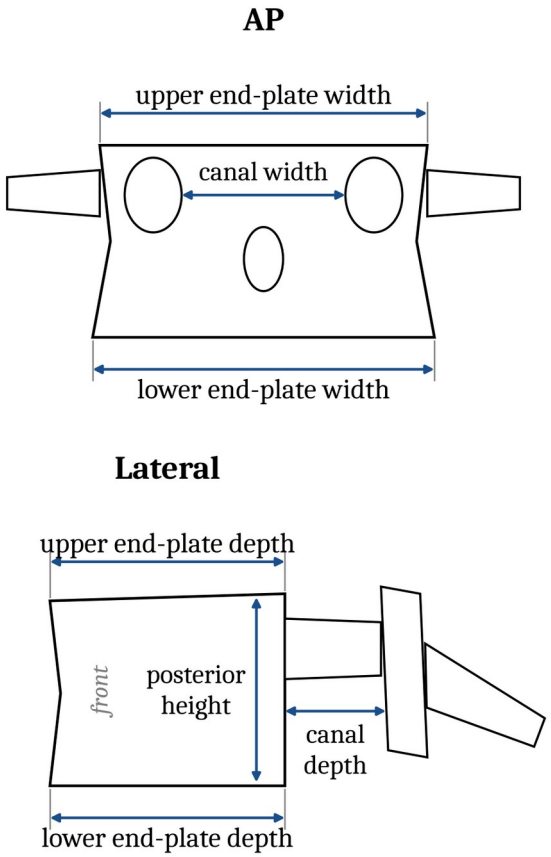
CNS 2026 ANNUAL MEETING

Evolution

Can the bodies in view name a vertebra?





Gregory Schwing, MD, PhD
General Surgery, Wayne State University School of Medicine / Detroit Medical Center

A view shows a few consecutive vertebral bodies, often with no C2 or sacrum to count from. Which levels are they?

$$x_{1:n} = (x_1, \ldots, x_n), \quad x_i \in \mathbb{R}^7, \quad k \in \{1, \ldots, 23 - n\}$$

Seven measures per body. The view's place *k* among the 22 levels from C3 to L5 names every body in it.

2,000 synthetic columns drawn from Panjabi's published means and spreads; every place of every column scored, so each rate below is over tens of thousands of fields.

Synthetic columns drawn from Panjabi's published means and spreads (Spine 1991;16:861, 888; 1992;17:299); no patient data. Retrospective modelling, 2026.

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Evolution

Bayes' rule over every place is the ceiling

$$p(k \mid x_{1:n}) = \frac{p(x_{1:n} \mid k)\, p(k)}{\sum_{k'=1}^{23-n} p(x_{1:n} \mid k')\, p(k')}$$

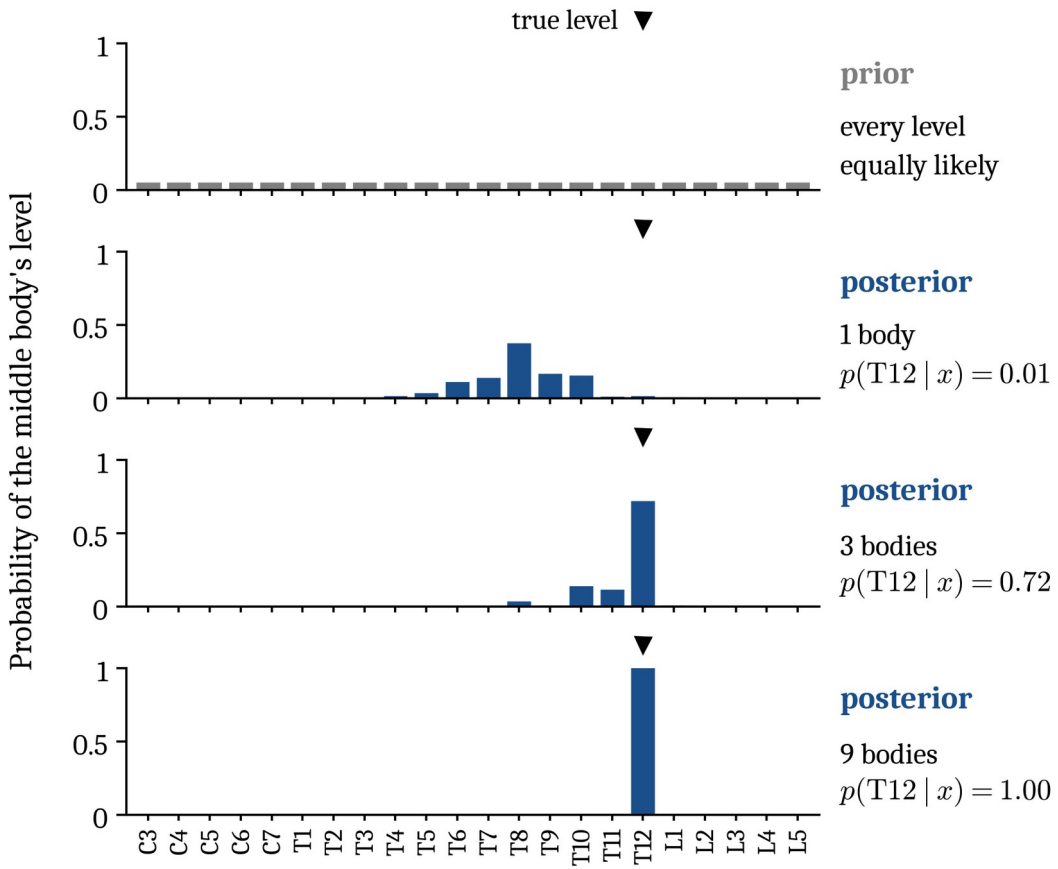
posterior ∝ likelihood × **prior**, over all 23 − n places

$$p(x_{1:n} \mid k) = \int \prod_{m=1}^{7} \mathcal{N}(x_{m,1:n} \mid s\,\mu_{m,k},\ \Sigma_{m,k}(s))\, p(s)\, ds$$

Every place is scored; nothing is fitted.

Given the patient's size *s*, each measure's values are Gaussian with a diagonal-plus-rank-one covariance, inverted in closed form; $\log s \sim \mathrm{N}(0, 0.07^2)$ is integrated by 161-node Gauss–Hermite quadrature.

So it is the Bayes rule for this generator: no formula or network on the same measures names more fields correctly on average.



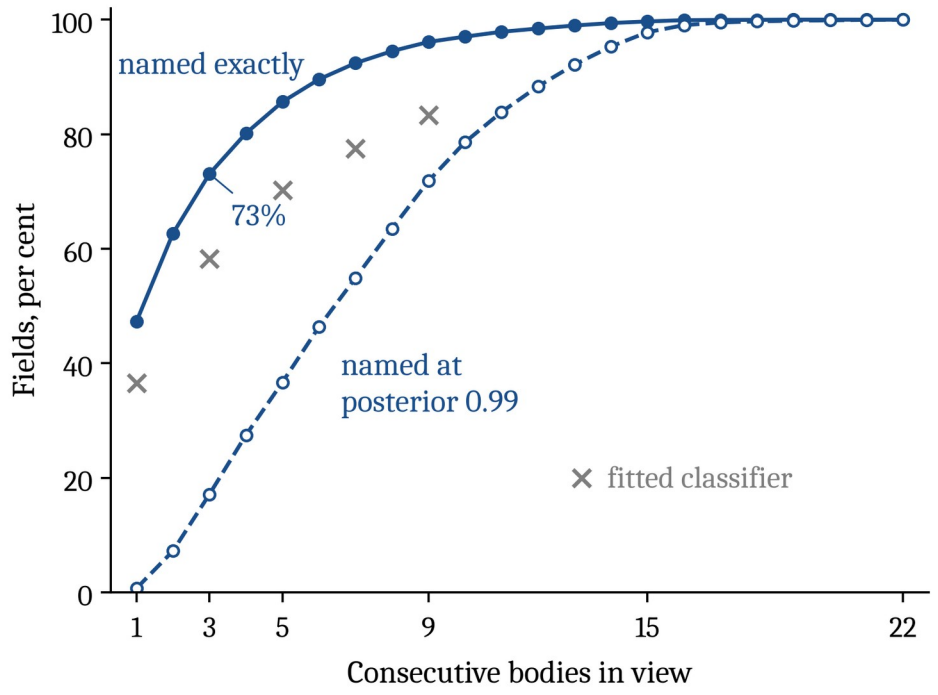
Synthetic columns drawn from Panjabi's published means and spreads (Spine 1991;16:861, 888; 1992;17:299); no patient data. Retrospective modelling, 2026.

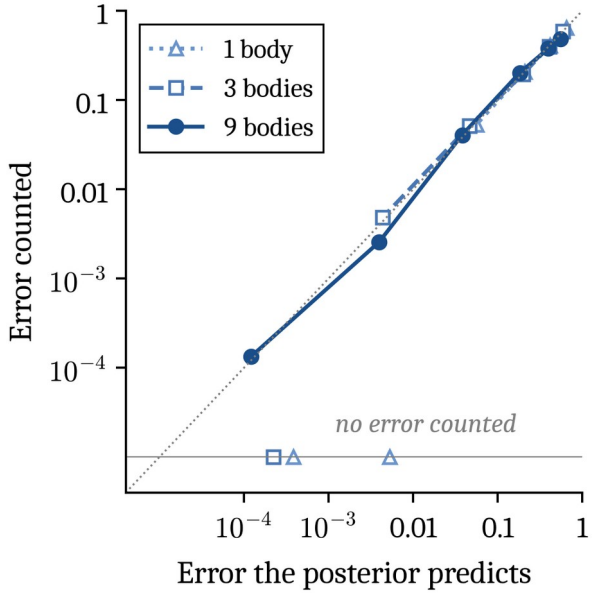
CNS

2026 ANNUAL MEETING

Evolution

Three bodies name 73% exactly; when sure, the name is right 99.8%





73.1%

of three-body fields named exactly (a fitted classifier: 58.2%)

99.78%

of names given at posterior 0.99 right (15 wrong of 6,848)

96.1%

named exactly from nine bodies

$$\Pr\left(\hat{k} = k \mid \hat{p} \geq 1 - \varepsilon\right) = \mathbb{E}[\hat{p} \mid \hat{p} \geq 1 - \varepsilon] \geq 1 - \varepsilon$$

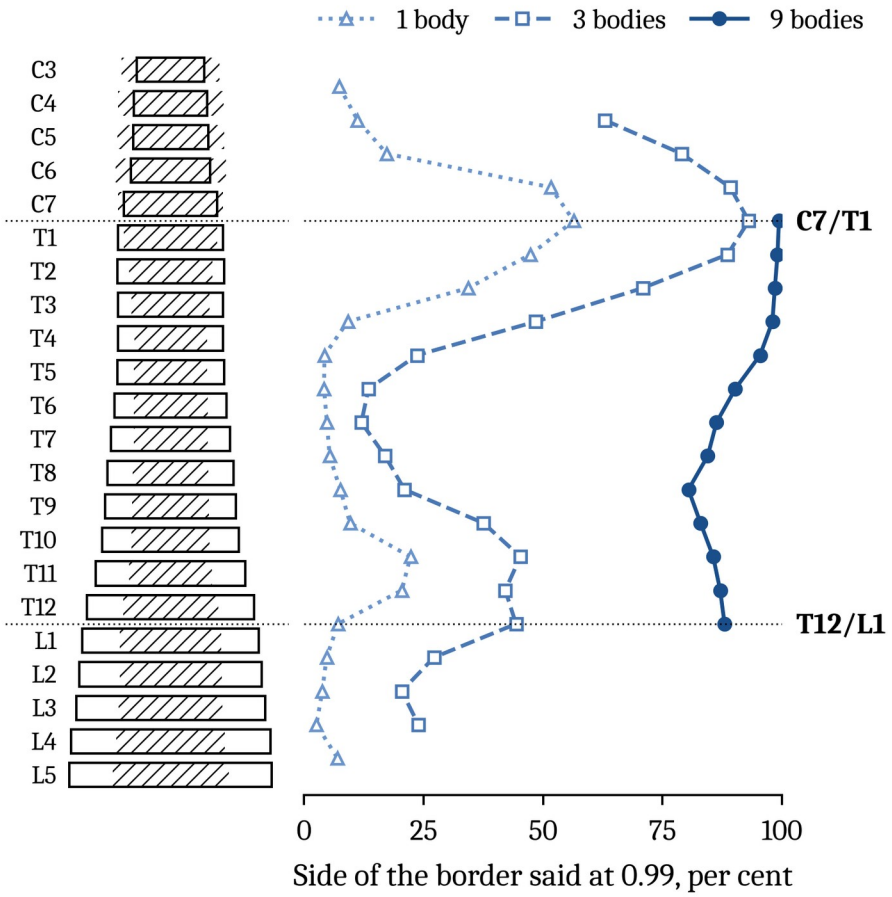
Answer only when the posterior $\hat{p}$ reaches 0.99, and the name is right at least 99% of the times it is given, while the generator is the truth.

Synthetic columns drawn from Panjabi's published means and spreads (Spine 1991;16:861, 888; 1992;17:299); no patient data. Retrospective modelling, 2026.

CNS
2026 ANNUAL MEETING

Evolution

One body tells its region and its side of C7/T1, rarely of T12/L1



$$\Pr(\ell \geq c \mid x_{1:n}) = \sum_{j \geq c} p(\ell = j \mid x_{1:n}) \geq 0.99$$

A border claim sums the same posterior, so a claim made at 0.99 is right at least 99% of the times it is made.

56%

of single bodies within two levels of C7/T1 say which side; right 99.84% (7 wrong of 4,514)

7%

at T12/L1, right 97.9% (12 wrong of 568): not near-certain

61%

of single bodies say their region; right 99.92% (21 wrong of 26,786), only 2 of them far from both junctions

88%

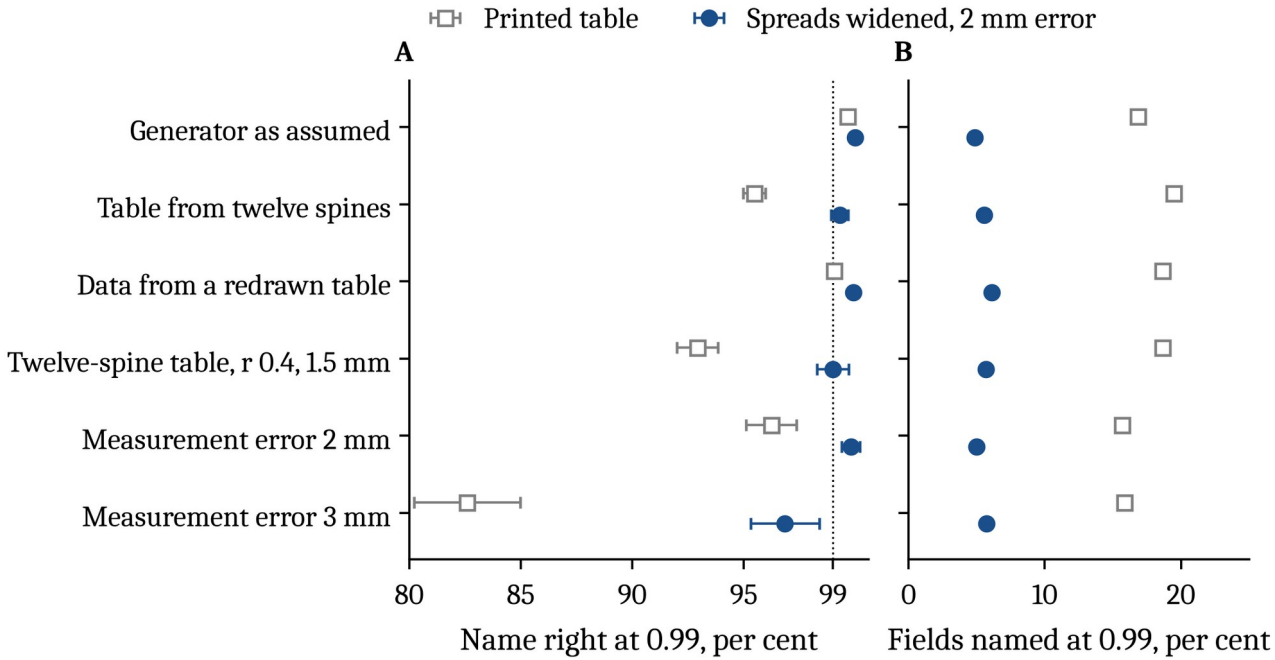
near T12/L1 say which side from nine bodies (1 wrong of 5,279)

Synthetic columns drawn from Panjabi's published means and spreads (Spine 1991;16:861, 888; 1992;17:299); no patient data. Retrospective modelling, 2026.

CNS 2026 ANNUAL MEETING

Evolution

The weak point is the table, not the mathematics



95.5%

of names at 0.99 right when the rule's table is estimated from 12 spines, as Panjabi's tables are

99.3%

with spreads widened and 2 mm of error, but 5.5% of fields named, not 19.5%, and lumbar names right 96.7%

$$\tilde{\sigma}_{m,\ell} = \sigma_{m,\ell}\sqrt{1 + 1/12}, \qquad \sigma_{\text{error}} = 2 \text{ mm}$$

Next: measure the table on full-spine CTs counted from C2; carry a real series' prior over places; add the anomalous columns (four or six lumbar bodies, a thirteenth rib-bearing body) as places of their own.

Synthetic columns drawn from Panjabi's published means and spreads (Spine 1991;16:861, 888; 1992;17:299); no patient data. Retrospective modelling, 2026.